Bow Valley College

# Course Outline

## PHRM1501

Hospital Pharmacy Practice

Winter 2022 - Fall 2022

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# PHRM1501 Hospital Pharmacy Practice

## COURSE DESCRIPTION

Learners will examine institutional pharmacy practice, including the role of hospital pharmacy within the health care system, organization of the dispensary, drug distribution systems, medication safety and current technology. Learners will be introduced to principles of aseptic technique for sterile production and quality assurance.

| | |
|---|---|
| REQUISITES | Complete all of the following<br>• Earn a minimum grade of C in each of the following courses<br>◦ PHRM1101 - Pharmacy Theory (3)<br>◦ PHRM1102 - Pharmacy Jurisprudence and Ethics (3)<br>• Complete with a minimum grade of C or concurrently enroll in all of the following courses<br>◦ PHRM1203 - Pharmacotherapy III (3)<br>• Complete with a minimum grade of A- or concurrently enroll in all of the following courses<br>◦ PHRM1302 - Pharmacy Calculations II (3) |
| EQUIVALENTS | None |
| CREDITS | 3 |
| HOURS | 45 |
| ELIGIBLE FOR PLAR | No |

## COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

**College-Wide Outcomes:**

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours

7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

| # | COURSE LEARNING OUTCOME(S) | COLLEGE WIDE OUTCOMES SUPPORTED |
|---|---|---|
| 1 | Examine hospital pharmacy within the health care system. | Outcome 2, 4, 7 |
| 2 | Examine hospital drug distribution systems and related policies and procedures. | Outcome 2, 4, 7 |
| 3 | Explore technology in hospital pharmacy practice. | Outcome 2, 5 |
| 4 | Examine medication safety best practices in the hospital setting. | Outcome 8, 10 |
| 5 | Apply principles of aseptic technique to the preparation of sterile products. | Outcome 1, 2, 3, 8 |
| 6 | Contribute to the effectiveness of working relationships in collaborative teams. | Outcome 1, 4, 9 |

## COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

| WEEK/HOURS | MODULES |
|---|---|
| Week 1 | Hospital Pharmacy Practice and the Health Care System Drug Distribution |
| Week 2 | Drug Distribution Prescribing and Dispensing Medication Technology in Hospital Pharmacy |
| Week 3 | Technology in Hospital Pharmacy Medication Safety |
| Week 4 | Sterile Production |
| Week 5 | Sterile Production |
| Week 6 | Review |
| Week 7 | Sterile Production |
| Week 8 | Midterm Exam |
| Week 9 | Sterile Production |
| Week 10 | Sterile Production |
| Week 11 | Sterile Production |
| Week 12 | Sterile Production |
| Week 13 | Sterile Production |
| Week 14 | Final Exam |

## ASSESSMENT

| COURSE LEARNING OUTCOME(S) | ASSESSMENT | WEIGHT |
|---|---|---|
| 1, 2, 3, 4, 5 | Assignments | 30% |
| 1, 2, 3, 4, 5 | Quizzes | 10% |
| 1, 2, 3, 4, 5 | Midterm Exam | 25% |
| 1, 2, 3, 4, 5 | Final Exam | 35% |

Important: For details on each assignment and exam, please see the Course Offering Information.

## PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

The Pharmacy Technician Program requires a minimum grade of C in this course to progress in the program.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

## GRADING SCHEME

| **Grade** | **Percentage** | **Grade Point** | **Description** |
|---|---|---|---|
| A+ | 95-100 | 4.0 | Exceptional: superior knowledge of subject matter |
| A | 90-94 | 4.0 | Excellent: outstanding knowledge of subject matter |
| A- | 85-89 | 3.67 | |
| B+ | 80-84 | 3.33 | |
| B | 75-79 | 3.0 | Very Good: knowledge of subject matter generally mastered |

| | | | |
|---|---|---|---|
| B- | 70-74 | 2.67 | |
| C+ | 67-69 | 2.33 | |
| C | 64-66 | 2.0 | Satisfactory/Acceptable: knowledge of subject matter adequately mastered |
| C- | 60-63 | 1.67 | |
| D+ | 57-59 | 1.33 | |
| D | 50-56 | 1.0 | Minimal Pass |
| F | Less than 50 | 0.0 | Fail: an unsatisfactory performance |

## REQUIRED LEARNING RESOURCES

Bow Valley College. (current year). *PHRM1501 Hospital pharmacy practice course package*.

Bow Valley College. (current year). *PHRM1501 Hospital pharmacy practice activity booklet*.

Johnston, M., Tsallas, G. *Fundamentals of Sterile Products Preparation for Pharmacy Technicians* (canadian ed.). Toronto, ON: Pearson Canada Inc.

Pickar, G., Pickar-Abernethy, A., Swart, B., Davis, M., O'Connell, J. (2021). *Dosage calculations* (5th ed.). Toronto, ON: Nelson

PHRM1501 Hospital Pharmacy Practice kit. Note: Available for purchase in bookstore.

Calculator (non-programmable).

Additional learning resources may be found in the Course Offering Information or in Brightspace.

## ADDITIONAL INFORMATION

**NAPRA Competencies:**

1.1 Practise within legal requirements.

1.1.1 Apply legal requirements to practice, including federal and provincial/territorial legislation, policies, by-laws, and standards.

1.1.2 Apply federal and provincial/ territorial workplace, occupational health and safety, and other related legislation to the practice setting.

1.1.3 Apply federal and provincial/territorial privacy legislation to the collection, use, storage, disclosure and destruction of personal health information.

1.2.1 Apply the principles of professional codes of ethics.

1.3.1 Identify illegal, unethical or unprofessional actions or situations.

1.3.2 Undertake appropriate intervention to address illegal, unethical or unprofessional actions or situations.

1.4.1 Apply principles of self-regulation.

1.4.2 Accept responsibility and accountability for own actions and decisions.

1.4.3 Seek guidance when uncertain about own knowledge, skills, abilities, and scope of practice.

1.4.4 Apply principles of continuing professional development including assessing own learning needs and developing a plan to meet these needs.

1.4.5 Maintain appropriate professional boundaries.

1.4.8 Describe the Canadian health care system and the role of health professionals within it.

3.1.1 Determine the validity, clarity, completeness and authenticity of the prescription and resolve concerns in collaboration with the pharmacist.

3.1.4 Interpret numerals, symbols, measurement systems and Latin abbreviations.

3.1.5 Perform pharmaceutical calculations.

3.1.6 Identify patterns of unusual drug prescribing and usage including possible diversion or drug misuse and report relevant findings to the pharmacist or appropriate authority.

3.2.7 Label products according to legislative requirements, best safety practices, established protocols and patient-specific needs.

3.3.1 Perform compounding calculations.

3.3.2 Prepare and compound sterile products according to recognized guidelines and standards of practice.

3.4.1 Identify when an independent double check should be performed.

4.1.1 Demonstrate the organizational and time management skills necessary to effectively prioritize, organize and manage product distribution workflow.

4.2.1 Apply inventory and formulary management systems and strategies that incorporate best practices, including new technologies.

5.2.3 Participate in organized initiatives for disaster, pandemic and emergency preparedness.

6.1.1 Clarify requests for information to identify questions that require pharmacist referral.

6.1.2 Use a variety of retrieval techniques to access reliable and appropriate information, including evidence-based information when possible.

6.2.2 Evaluate the information and use current, relevant and reliable information to improve practice.

7.1.1 Demonstrate proficiency in written and verbal English or French.

7.1.2 Demonstrate appropriate verbal and non-verbal communication skills, including listening skills.

7.1.5 Conduct interpersonal interactions, including conflict management, in a professional manner.

7.1.6 Communicate with sensitivity, respect and empathy.

8.1.1 Identify potential collaborators with whom to initiate ongoing professional relationships.

8.1.2 Collaborate with other parties in the relationship to define the roles and responsibilities of each party.

8.2.1 Interact respectfully with other members of the team by accepting accountability for themselves and managing disagreements and conflict.

8.3.1 Collaborate with team members to ensure appropriate utilization of resources.

8.4.1 Recognize situations that fall beyond the scope of practice of pharmacy technicians and refer these situations to the pharmacist.

to a culture of patient safety.

9.1.1 Apply principles of patient safety to improve practice.

9.2.4 Identify high-alert drugs and high-risk processes in order to respond effectively.

9.3.1 Maintain the cleanliness, functionality and integrity of compounding, packaging, dispensing, and storage equipment.

9.3.2 Ensure that products are stored and transported under the conditions required to maintain product quality, safety and integrity, including cold chain management.

9.4 Create and maintain a working environment that promotes safety.

9.4.3 Identify factors that impact the safety of the working environment, including resource allocation, procedural consistency and ergonomics.

Additional information may be found in the Course Offering Information or in Brightspace.

## ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

## INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

**Turnitin:**

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

**Online Exam Proctoring:**

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an

exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

**Collection of Personal Information:**

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.